

TerraFly — Critical Fix Specification

3D Rendering System and Height Estimation System — Mandatory Requirements

Problem statement	SIH26175 – DepthWizard: Single-View Height Estimation and 3D Flythrough
Sponsor	ISRO / Department of Space, Smart India Hackathon 2026
Deadline	20 September 2026
Document purpose	Mandatory fix specification for the two currently failing subsystems: 3D rendering output and height/depth e
Audience	Engineering agent (Codex) implementing the fixes, and the TerraFly team reviewing the result against the

“A single photo can tell you shape. It cannot, by itself, tell you real-world metres. Every requirement in this document exists in service of that one fact being handled correctly and shown honestly.”

This document supersedes prior informal fix requests. Implement every requirement in Sections 2 and 3. Do not mark this complete until every row in the Section 6 acceptance checklist passes.

0. Standing Rule — Applies to Every Requirement Below

Relative height (unitless shape) and absolute/metric height (real metres, calibrated against a reference) are two different outputs with two different confidence levels. They must never be computed, stored, displayed, or discussed as if they were the same kind of number. Every fix in this document must preserve this separation. If a change is proposed that would make it harder to tell which mode produced a given value, reject that change and find another approach.

1. Current Failure State

The following was confirmed by direct inspection of an exported model file and rendered output, not by assumption. Treat both as verified defects, not stylistic complaints.

Subsystem	Observed defect	Evidence
3D rendering – texture	Terrain colour is baked as one flat colour per mesh vertex (192×192 grid) with an unlit material. No image texture, no UV coordinates exist in the export. Output looks blurred/blocky regardless of source photo resolution.	glTF inspection: zero images/textures in file; COLOR_0 attribute present; KHR_materials_unlit set.
3D rendering – shading	No NORMAL attribute exists on the mesh. Lighting cannot render correctly even after the texture fix above, because there is no surface-direction data for a renderer to shade with.	glTF inspection: primitive attributes = POSITION, COLOR_0 only.
3D rendering – geometry	Height field is noisy at pixel level. Rooftops render as rounded domes instead of flat planes; building edges render as smooth slopes instead of sharp transitions.	Visual inspection of rendered output: uniformly rounded building forms, no flat roof planes.
3D rendering – texture on steep faces	Steep mesh faces (building sides) still receive a stretched single-row sample of the top-down photo, producing visible vertical drip/streak artefacts.	Visual inspection: elongated vertical colour streaks on every raised structure.
Height model	Backbone in use is the smallest, least accurate public checkpoint of a general-purpose depth model not trained on remote-sensing imagery. No fine-tuning, no benchmark evidence.	Model identifier confirmed as Depth Anything V2 “Small”. No metric card or benchmark artefact exists in the project.

2. Rendering System — Mandatory Fixes

2.1 Real texture, not baked vertex colour

Remove COLOR_0-based colouring and KHR_materials_unlit entirely. Generate TEXCOORD_0 (UV) coordinates for every vertex, mapped across the mesh footprint in the same orientation as the height grid. Embed the source photo as a real glTF image and texture at native resolution, capped at 2048×2048 only if file size becomes a genuine constraint. Bind it as baseColorTexture on a standard pbrMetallicRoughness material. Mesh vertex density and texture resolution are independent of each other from this point forward — do not couple them.

2.2 Real shading, not flat unlit colour

Compute smooth per-vertex normals from the height field (central-difference gradient of neighbouring heights, or averaged adjacent face normals) and write them as the NORMAL accessor. Add one directional light and one

low-intensity ambient light to the viewer scene. Verify visually that identical geometry looks different under different simulated sun angles — if it does not, the normals or the lighting setup are still wrong.

2.3 Edge-preserving height smoothing before meshing

Apply a guided/joint bilateral filter to the raw height map before mesh generation, using the RGB image as the guidance signal. A plain Gaussian blur is not acceptable — it will soften real building edges along with the noise. After filtering, run a rooftop-flattening pass: segment near-constant-height connected regions and replace each region's height with its median or a fitted-plane value so rooftops present as flat surfaces. Clip single-pixel height outliers that sit far outside their local neighbourhood's range before meshing.

2.4 Distinct handling for steep (wall) faces

Detect mesh faces above a defined slope threshold. These are wall faces, not roof or ground faces, and must not sample the stretched top-down photo. Give them a synthetic material: flat neutral shade or a subtle vertical gradient. Roof and ground faces keep the real photo texture. If time allows, treat this as a stretch goal beyond the minimum: instead of one continuous heightfield mesh, detect discrete rooftop regions and extrude each as a flat-roofed block with separate wall geometry — closer to standard LOD1 building-model generation and the most robust way to eliminate this defect class entirely.

2.5 On-screen height legend (colour scale)

Every relative and calibrated height view must display a visible colour-scale legend in a fixed screen corner, mapping colour directly to height. Use a two-colour gradient (for example green at the low end through red at the high end) with numeric tick labels at minimum, midpoint, and maximum. The legend is not decorative — a viewer must be able to look at any coloured pixel in the scene and read its approximate height straight off the legend without hovering or clicking.

The two view modes require different scaling behaviour and must not be confused:

View mode	Legend scaling rule
Relative / uncalibrated (PNG, JPG)	Scale the legend to that scene's own minimum and maximum raw value. There is no absolute reference in this mode, so per-scene auto-scaling is correct and expected. Label the legend “relative – not metres” so it is never mistaken for the calibrated mode.
Calibrated / metric (GeoTIFF, GCP or SRTM fitted)	Scale the legend in real metres using the fitted calibration line from Section 3.5. Prefer a consistent fixed metre range across demo scenes where practical, so colours are comparable scene to scene rather than re-normalised every time. Label the legend with its unit (“metres”) explicitly.

2.6 Explicit prohibition

Do not respond to any of the above by simply increasing mesh grid resolution. Higher resolution on an unsmoothed, un-textured height field produces more individual noisy spikes and more stretched streaks, not fewer. Resolution is not a substitute for correctness in Sections 2.1–2.4.

3. Height Estimation System — Mandatory Upgrade Path

3.1 Immediate step — checkpoint upgrade (no training required)

Replace the Depth Anything V2 “Small” checkpoint with “Large”. This is a same-day configuration change on the existing code path, not a rebuild. Re-run the existing test image set and confirm visually reduced noise before proceeding to 3.2. Do not skip this step even if 3.2 is planned — it is the fastest available quality improvement and must be verified independently.

3.2 Domain adaptation — do not skip

A general-purpose depth model was trained on ground-level photography, not overhead remote-sensing imagery. This is the primary source of remaining error and must be addressed directly, not worked around. Two published approaches are directly relevant and must be evaluated before any custom architecture work begins:

Reference	What it is	Required action
RDAH-Net – “Bridging Relative Depth and Absolute Height for Monocular Height Estimation in Remote Sensing,” <i>MDPI Remote Sensing</i> , March 2026	A published architecture built specifically for this relative-to-absolute height problem in remote sensing.	Search for an official code/weights release before committing engineering time to a from-paper reimplementation. If none exists, do not reimplement the full architecture under the current deadline — treat as a stretch goal only.
Depth2elevation – IEEE <i>Transactions on Geoscience and Remote Sensing</i>	Adapts Depth Anything specifically for single-view remote-sensing height estimation via scale modulation, rather than replacing the backbone.	Prefer this path if a lower-risk integration is needed — it modifies the existing backbone instead of requiring a new architecture.

3.3 Fine-tuning data

Use the ISPRS Potsdam and Vaihingen benchmark datasets (real aerial RGB paired with real LiDAR-derived DSM ground truth). Verify and record licence terms before use. Hold out a portion of each dataset strictly for evaluation — never train on the held-out split. This held-out split is also the source of the Section 3.6 benchmark and must not be reused for fitting anything.

3.4 Hardware reality check — do not assume unlimited capacity

The available hardware is an RTX 5060 with 8 GB VRAM, a Ryzen 5 7600, and 16 GB system RAM. This is sufficient for inference at moderate tile sizes and for parameter-efficient fine-tuning. It is not assumed to be sufficient for full-precision full fine-tuning of the Large checkpoint without the mitigations below. Do not attempt full fine-tuning at default settings and discover an out-of-memory failure after hours of setup — plan for the constraint up front.

Constraint	Required mitigation
8 GB VRAM during fine-tuning	Use mixed precision (fp16/bf16), gradient checkpointing, and small batch size (1–2) with crop-based training rather than full-scene batches.
Full fine-tune may still not fit or may overfit on a modest dataset	Default to parameter-efficient fine-tuning (freeze most backbone layers; fine-tune the final layers plus the depth head, or use LoRA-style adapters). Treat full fine-tuning as an optional stretch goal only if VRAM and time both allow it.

Constraint	Required mitigation
Time budget is fixed (deadline 20 September 2026)	If fine-tuning is not converging usefully within its allotted time window, fall back to training a lightweight calibration head on top of the frozen backbone's features. This directly serves the existing Calibrate pipeline stage and is far cheaper to get right.
16 GB system RAM with large rasters/meshes	Tile or downsample large inputs during preprocessing; release intermediate arrays explicitly rather than relying on garbage collection timing.

3.5 Minimum reference points for calibration

The relative-to-metric conversion is a linear fit: $\text{real height} = (a \times \text{raw model value}) + b$. This has two unknowns (scale a , offset b), so a minimum of two spatially distinct known reference points is mathematically required — one known point alone cannot determine both. In practice, fit the line against a minimum of five to twenty scattered reference points (SRTM samples and/or GCPs) using a robust method such as RANSAC or least-squares with outlier rejection, not a simple two-point solve, because the raw model output is not perfectly consistent across a whole scene. Never present a metric height derived from a single reference point as reliable — the calibrated-view badge in the results panel must reflect how many reference points were actually used.

3.6 Mandatory benchmark output

After any model change (checkpoint swap, fine-tune, or domain-adaptation method), run inference on the held-out validation split and compute RMSE, MAE, and Pearson correlation against real reference elevation, using only valid (non-NoData) pixels present in both prediction and reference. Record results in a metric card: dataset and licence, exact checkpoint/commit, split definition, coordinate system and resolution, the three metrics, runtime and peak VRAM/RAM, and any landscape category with no available ground truth stated honestly rather than omitted or estimated. Do not report a number that was not computed from real reference data.

3.7 Explicit prohibition

Do not attempt to solve height/depth prediction quality with a general-purpose large language model API (for example a chat-oriented API such as Grok or a locally hosted LLM such as Llama). These are text models and cannot perform pixel-wise depth regression. Any budget or API-key decision for this project must be spent on vision/depth-specific models and tooling, not conversational LLM APIs.

3.8 Optional: shadow-based independent cross-check

As a validation aid, not a replacement for the model pipeline: for isolated structures with a clearly visible, unambiguous shadow, compute height independently via $\text{height} = \text{shadow length} \times \tan(\text{sun elevation angle})$, where the sun elevation angle is derived from the image's capture timestamp and location (standard solar-position formulas), and shadow length is measured directly in the image. This method uses no camera-altitude data and no AI model — it is pure geometry. Restrict its use to a small number of clean test cases (isolated buildings, unambiguous shadow, known capture time). It will fail or be unreliable in overcast conditions, near solar noon, in dense urban blocks where shadows overlap or fall on other structures, and under forest canopy. Where it succeeds, report it in the metric card as an independent, non-ML sanity check alongside the Section 3.6 benchmark — this is a credibility asset in front of evaluators, not a core deliverable.

4. Interaction Between Sections 2 and 3

Complete Section 2 before evaluating Section 3 results. An improved height model viewed through a broken renderer cannot be visually judged — noisy geometry and model noise look identical on screen once passed through the current unlit, un-smoothed, un-textured pipeline. Confirm Section 2's acceptance criteria pass on the existing model first, then re-benchmark after each Section 3 change so rendering quality and model quality are never conflated when deciding whether a change worked.

5. Presentation Note — Not a Code Change, Read Before the Demo

Natural terrain (hills, open landscape) renders convincingly through this pipeline because the surface is genuinely continuous — there is no missing geometry to invent, so the current approach is close to physically correct there already. Buildings are the harder case: a top-down photograph never shows a wall, only a roof, so any building height shown is an informed inference, not a measurement of something the camera could see. This is a property of the problem, not a flaw in the implementation.

Recommended framing for the live demonstration: lead with a natural-terrain result as the primary visual — it is the strongest, most physically defensible output the system currently produces. Follow with an urban example and state the wall-inference limitation before a judge asks about it. Close with the Section 3.6 benchmark table as evidence. Volunteering a known limitation reads as more credible to an evaluator than waiting to be asked and improvising an answer.

6. Acceptance Checklist — All Rows Must Pass

#	Requirement	Pass condition
1	Texture	Rendered surface shows the real source photo at full sharpness via UV mapping, not a flat baked colour per vertex.
2	Shading	Identical geometry visibly changes appearance when the simulated light direction is changed.
3	Roof geometry	A building rooftop in the rendered scene is a flat plane, not a dome or rounded peak.
4	Wall faces	No stretched vertical colour streaking is visible on any raised structure.
5	Model checkpoint	Depth Anything V2 Large (or better) is confirmed as the active checkpoint, not Small.

#	Requirement	Pass condition
6	Domain adaptation	RDAH-Net and Depth2elevation have been checked for a public code/weights release, with the outcome documented either way.
7	Benchmark evidence	A metric card exists with real RMSE, MAE, and Pearson correlation computed against held-out reference data, not estimated.
8	Mode labelling	Every displayed height value is visibly labelled as either calibrated/metric or relative/uncalibrated — never ambiguous.
9	Hardware fit	Fine-tuning runs to completion on the available 8 GB VRAM without an out-of-memory failure.
10	Height legend	A colour-scale legend with numeric tick labels is visible on screen for both relative and calibrated views, scaled per Section 2.5.

This specification does not authorise any new feature outside Sections 2 and 3. Complete and verify all ten checklist rows before resuming other roadmap work.